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BATTERY CELL INSPECTION SYSTEM

Abstract

A battery cell inspection system includes a transport unit including a plurality of conveyors arranged in multiple levels in a vertical direction, and configured to transport a tray on which an assembled battery cell is mounted, a cell inspection unit corresponding to the multiple levels of the transport unit, and configured to perform pre-charging and cell defect inspection of the battery cell, and a shuttle configured to move between the cell inspection unit and the transport unit to carry the tray into the cell inspection unit or take the tray from the cell inspection unit.

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Background/Summary

TECHNICAL FIELD

[0001] The present application claims priority to Korean Patent Application No. 10-2022-0070545 filed on Jun. 10, 2022 in the Republic of Korea, the disclosures of which are incorporated herein by reference.

[0002] The present disclosure relates to a battery cell inspection system, and more particularly, to a battery cell inspection system with improved spatial density and battery production efficiency by improving a logistics movement path and an inspection path in a formation process for activation and operation inspection of an assembled battery cell.

BACKGROUND ART

[0003] Recently, there has been a rapid increase in the demand for portable electronic products such as laptop computers, video cameras and mobile phones, and with the extensive development of electric vehicles, accumulators for energy storage, robots and satellites, many studies are being made on high performance secondary batteries that can be recharged repeatedly.

[0004] Currently, commercially available secondary batteries include nickel-cadmium batteries, nickel-hydrogen batteries, nickel-zinc batteries, lithium ion secondary batteries and the like. Among them, lithium ion secondary batteries have little or no memory effect, and thus they are gaining more attention than nickel-based secondary batteries for their advantages that recharging can be done whenever it is convenient, the self-discharge rate is very low and the energy density is high. The lithium ion secondary batteries primarily include lithium-based oxides and carbon materials for a positive electrode active material and a negative electrode active material, respectively. The assembly of the lithium ion secondary batteries is performed by receiving an electrode assembly in a case together with an electrolyte material, wherein the electrode assembly includes a positive electrode plate and a negative electrode plate coated respectively with the positive electrode active material and the negative electrode active material and a separator interposed between the positive electrode plate and the negative electrode plate.

[0005] Meanwhile, the assembled battery cell undergoes a formation process for cell activation and operation inspection.

[0006] However, according to the conventional process, cell activation and operation inspection is performed while a single level tray on which assembled battery cells are mounted is primarily being transported using a conveyor, resulting in low spatial density, long inspection time and slow inspection speed, which increases the takt time of the formation process and reduces the battery cell production efficiency.

SUMMARY

Technical Problem

[0007] The present disclosure is designed to solve the above-described problem, and therefore the present disclosure is directed to providing a battery cell inspection system having an improved structure of receiving a plurality of assembled battery cells transported, and simultaneously carrying a plurality of trays to and from a cell inspection unit, thereby improving the spatial density and the production efficiency of the entire battery inspection system.

[0008] However, the technical problem to be solved by the present disclosure is not limited to the above-described problems, and these and other problems will be clearly understood by those skilled in the art from the following description.

Technical Solution

[0009] A battery cell inspection system according to an aspect the present disclosure may include a transport unit including a plurality of conveyors arranged in multiple levels in a vertical direction, and configured to transport a tray on which an assembled battery cell is mounted, a cell inspection unit corresponding to the multiple levels of the transport unit, and configured to perform pre-charging and cell defect inspection of the assembled battery cell, and a shuttle configured to move between the cell inspection unit and the transport unit to carry the tray into the cell inspection unit or take the tray from the cell inspection unit.

[0010] The shuttle may be configured to convey a plurality of trays when carrying the plurality of trays to the cell inspection unit or when taking the plurality of trays from the cell inspection unit.

[0011] The shuttle may include multiple levels corresponding to the transport unit and the cell inspection unit, and wherein each level of the multiple levels of the shuttle may be configured to convey at least one tray.

[0012] The transport unit may be disposed on a carry-in path and a carry-out path of the cell inspection unit.

[0013] The transport unit may include a first transport conveyor, and a second transport conveyor disposed a vertical distance from the first transport conveyor.

[0014] The cell inspection unit may further include a first inspection device disposed at a height corresponding to the first transport conveyor and a second inspection device placed on the first inspection device and disposed at a height corresponding to the second transport conveyor.

[0015] Each of the first inspection device and the second inspection device may include a pin plate disposed on the tray and having a plurality of exposed electrodes, a bottom plate configured to support the tray and spaced a predetermined distance apart from the pin plate, and an up/down movement unit connecting the bottom plate to the pin plate, and configured to decrease or increase a distance between the bottom plate and the pin plate.

[0016] Each of the first inspection device and the second inspection device may further include a jig unit disposed on the bottom plate and configured to support and align the tray.

[0017] The up/down movement unit may include a cylinder, a rod having an end connected to the cylinder and an opposite end coupled to the bottom plate, and a fixing block coupled to the bottom plate and configured to couple the opposite end of the rod to the bottom plate.

[0018] The jig unit may include a jig plate configured to receive the tray, and a ratchet fastener configured to fix two sides of the tray received on the jig plate when the ratchet fastener is rotated.

[0019] The jig unit further may include a stopper configured to limit movement in a direction of carrying the tray into the jig plate.

[0020] A plurality of cell inspection units may be arranged in a line.

[0021] The shuttle may move in parallel to a lengthwise arrangement of the plurality of cell inspection units.

[0022] The shuttle may include a body having a moving wheel, a first carrying unit disposed at the height corresponding to the first transport conveyor and the first inspection device, and a second carrying unit disposed the height corresponding to the second transport conveyor and the second inspection device.

[0023] Each of the first carrying unit and the second carrying unit may include a pair of guide rails spaced apart from each other by a width corresponding to a width of the tray, a fork for gripping that is configured to slide along the pair of guide rails and configured to grip two sides of the tray, and a fork driving unit configured to drive the fork for gripping.

[0024] The fork for gripping may move from the body in a direction perpendicular to a movement direction of the shuttle.

[0025] The cell defect inspection may include causing a predetermined amount of current to flow in the assembled battery cell, detecting a voltage change of the assembled battery cell and determining a presence or absence of a defect.

[0026] The assembled battery cell may include an electrode assembly including a negative

electrode plate, and the pre-charging of the assembled battery cell may include charging the assembled battery cell in a predetermined amount to form an electric charge layer on the negative electrode plate to prevent formation of an electron movement channel by oxidation of metal impurities.

[0027] The assembled battery cell may be a cylindrical battery.

Advantageous Effects

[0028] According to an aspect of the present disclosure, it may be possible to form an improved structure of receiving a plurality of assembled battery cells transported, and simultaneously carrying a plurality of trays to the cell inspection unit and taking the plurality of trays from the cell inspection unit, thereby improving the spatial density and the production efficiency of the entire battery inspection system.

[0029] Additionally, the plurality of trays may be simultaneously inspected by the cell inspection unit that performs cell defect inspection and pre-charging, so it may be possible to reduce the space occupied by the inspection machine, thereby improving the space efficiency of the battery inspection system.

[0030] Additionally, according to an aspect of the present disclosure, the plurality of trays on which the battery cells are mounted may be simultaneously carried to and from the cell inspection unit each time, so it may be possible to reduce the inspection time and improve the inspection speed, thereby reducing the takt time of the battery cell formation process and increasing the battery cell production efficiency.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0031] The accompanying drawings illustrate exemplary embodiments of the present disclosure and together with the following detailed description, serve to provide a further understanding of the technical aspect of the present disclosure, and thus the present disclosure should not be construed as being limited to the drawings.

[0032] FIG. 1 is a plan view showing the layout of a battery cell inspection system according to an embodiment of the present disclosure.

[0033] FIG. 2 is a diagram schematically showing a battery inspection system according to an embodiment of the present disclosure.

[0034] FIG. 3 is a perspective view of a tray having battery cells that will be supplied to a battery cell inspection system according to an embodiment of the present disclosure.

[0035] FIG. 4 is a perspective view of a cell inspection unit of a battery inspection system according to an embodiment of the present disclosure.

[0036] FIG. 5 is a front view of a cell inspection unit of a battery inspection system according to an embodiment of the present disclosure.

[0037] FIGS. 6 and 7 are top views of a bottom plate and a jig unit of a cell inspection unit of a battery inspection system according to an embodiment of the present disclosure.

[0038] FIG. 8 is a perspective view of a shuttle of a battery inspection system according to an embodiment of the present disclosure.

[0039] FIGS. 9 and 10 are diagrams showing a tray lifted by a fork for gripping and a protruding portion of a shuttle of a battery inspection system according to an embodiment of the present disclosure.

[0040] FIGS. 11 to 13 are diagrams showing an example in which a cell inspection unit of a battery inspection system according to an embodiment of the present disclosure accommodates trays from a transport unit and performs cell defect inspection and pre-charging.

DETAILED DESCRIPTION

[0041] Hereinafter, exemplary embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. Prior to the description, it should be understood that the terms or words used in the specification and the appended claims should not be construed as being limited to general and dictionary meanings, but rather interpreted based on the meanings and concepts corresponding to the technical aspect of the present disclosure on the basis of the principle that the inventor is allowed to define the terms appropriately for the best explanation.

[0042] Therefore, the embodiments described herein and the illustrations shown in the drawings are exemplary embodiments of the present disclosure to describe the technical aspect of the present disclosure and are not intended to be limiting, so it should be understood that a variety of other equivalents and modifications could have been made thereto at the time that the application was filed.

[0043] In the drawings, each element or a specific portion of the element may be exaggerated, omitted or schematically depicted for convenience of description and clarity. Accordingly, each element may not be shown at true scale size. When it is determined that a certain detailed description of related known functions or elements may unnecessarily obscure the subject matter of the present disclosure, the description is omitted.

[0044] FIG. 1 is a plan view showing the layout of a battery cell inspection system according to an embodiment of the present disclosure, FIG. 2 is a diagram schematically showing a battery inspection system according to an embodiment of the present disclosure, FIG. 3 is a perspective view of a tray having battery cells that will be supplied to the battery cell inspection system according to an embodiment of the present disclosure, and FIG. 4 is a perspective view of a cell inspection unit of the battery inspection system according to an embodiment of the present disclosure.

[0045] Referring to FIGS. 1 to 3, the battery cell inspection system according to an embodiment of the present disclosure includes a tray **30** on which a battery cell **20** to be inspected is mounted, a transport unit **100** to transport the tray **30**, the cell inspection unit **200** receives and inspects the battery cell **20**, and a shuttle **300** to convey the tray **30** to the transport unit **100** and the cell inspection unit **200**.

[0046] The object to be inspected by the battery inspection system according to an embodiment of the present disclosure will be described. The battery cell **20** may be a secondary battery, for example, a pouch type secondary battery, a prismatic secondary battery or a cylindrical secondary battery. Hereinafter, this embodiment will be described based on cylindrical secondary batteries as the battery cells **20** as shown in FIG. 3.

[0047] The tray **30** may be where the assembled battery cells **20** are mounted. The tray **30** may be approximately rectangular or square in shape when viewed from the top, and may accommodate the plurality of battery cells **20**. The tray **30** may have seating grooves **32** (see FIG. 13) in which the battery cells **20** are seated. The seating groove **32** may support and secure each battery cell **20**.

[0048] The number of battery cells **20** mounted in the tray **30** may vary, and in the drawing, the number of battery cells **20** that may be mounted in a tray may be 8*8, 64. The height of the battery cells **20** mounted in the tray **30** may be lower than the top of the tray **30** and may be at the level of 80 to 90% of the thicknesswise height of the tray **30**.

[0049] Referring primarily to FIGS. 1 and 2, the transport unit **100** may transport the tray **30** in which the assembled battery cells **20** are mounted. The transport unit **100** may be disposed on a carry-in path and a carry-out path of the cell inspection unit **200**. That is, the transport unit **100** may be configured to transport the tray **30** from an assembly unit **5** to a location for taking the tray **30** to the cell inspection unit **200**, and the transport unit **100** may be configured to transport the tray **30** from a location at which the tray **30** is taken from the cell inspection unit **200** to a post-process (for example, an aging machine **9** or a charging/discharging process). Meanwhile, a station **11** (or a lift) may be disposed between the assembly unit **5** and the transport unit **100** as shown in FIG. 1.

[0050] The transport unit **100** may include a plurality of conveyors arranged in multiple levels in

the vertical direction. The transport unit **100** transports each tray **30** in a single level, and may have a multistorey, multilevel structure for simultaneously transporting the trays **30** at different heights. To this end, the transport unit **100** may include the plurality of conveyors arranged in parallel in the vertical direction.

[0051] Hereinafter, this embodiment will be described based on two conveyors and accordingly, the cell inspection unit **200** and the shuttle **300** in two levels.

[0052] Referring primarily to FIG. 2, the transport unit **100** may include a first transport conveyor **110**, and a second transport conveyor **120** disposed vertically above the first transport conveyor **110**. A single level tray **30** may be loaded on each of the first transport conveyor **110** and the second transport conveyor **120**.

[0053] Meanwhile, at least one (one tray **30** in FIG. 2) single level tray **30** may be disposed on each of the first transport conveyor **110** and the second transport conveyor **120**, and may change by adjusting the transport length of the first transport conveyor **110** and the second transport conveyor **120**. That is, when the length of the conveyor increases, the length of the first transport conveyor **110** and the second transport conveyor **120** is longer than that of FIG. 2, and in this case, the number of trays **30** placed in each transport conveyor **110**, **120** may increase.

[0054] Due to the double level structure of the transport unit **100** in this embodiment, it may be possible to improve the process of carrying the tray **30** to and from the cell inspection unit **200**. That is, two trays **30**, each loaded on each of the first transport conveyor **110** and the second transport conveyor **120**, may be simultaneously carried to the cell inspection unit **200** or taken from the cell inspection unit **200** to the transport unit **100**. Accordingly, the longer conveyor may be used as a buffer space in which the trays **30** accumulate on the conveyor, thereby minimize the reduction in space utilization.

[0055] Further, compared to one conveyor, each one tray **30** on each of the first transport conveyor **110** and the second transport conveyor **120**, totaling two trays **30** may be simultaneously carried to or from the cell inspection unit **200**, so it may be possible to reduce the inspection time and improve the inspection speed, thereby improving the overall battery production efficiency.

[0056] Hereinafter, the formation process for activation and operation inspection of the assembled battery cells **20** will be briefly described. The formation process largely includes pre-charging/low current inspection of the battery cells **20**, followed in a sequential order by, aging through a loaded item **7** loaded in multilayer and the aging machine **9**, and charging/discharging. Here, pre-charging and low current inspection may be performed at the early step of the formation process.

[0057] The pre-charging may prevent the formation of an electron movement channel by oxidation of metal impurities by charging the battery cells **20** in a predetermined amount to form an electric charge layer on the negative electrode plate. Additionally, the cell defect inspection may determine the presence or absence of defects by detecting a voltage change of the battery cells **20** after the flow of the predetermined amount of current in the battery cells **20**. Through the cell inspection unit **200**, the pre-charging and low current inspection process may be performed.

[0058] The cell inspection unit **200** may be a device that performs cell defect inspection and pre-charging of the battery cells **20**. Referring primarily to FIG. 1, a plurality of cell inspection units **200** may be arranged in a line. In the drawing, six cell inspection units are arranged in the up-down direction and two inspection lines are symmetrically arranged. Additionally, buffer units **130**, **140** may be disposed on one side of the cell inspection unit **200**.

[0059] Referring primarily to FIG. 4, the cell inspection unit **200** may include a first inspection device **210** disposed at a height corresponding to the first transport conveyor **110**; and a second inspection device **220** placed on the first inspection device **210** and disposed at a height corresponding to the second transport conveyor **120**.

[0060] The cell inspection unit **200** may be a double level corresponding to the first transport conveyor **110** and the second transport conveyor **120**. Accordingly, the first inspection device **210** and the second inspection device **220** may be disposed in a lower area and an upper area within a

frame **201** and vertically stacked on top of each other. The height to which the first inspection device **210** and the second inspection device **220** accommodate the tray **30** may correspond to the height to which the tray **30** is seated in the first transport conveyor **110** and the second transport conveyor **120**, respectively.

[0061] The first inspection device **210** and the second inspection device **220** may have the same structure. That is, each of the first inspection device **210** and the second inspection device **220** may include a pin plate **230** having exposed electrodes **231**, a bottom plate **240** configured to support the tray **30**, and an up/down movement unit **260** connecting the bottom plate **240** to the pin plate **230**.

[0062] A power unit **202** may be placed on top of the lower area and the upper area of the frame **201**. The power unit **202** may supply power to the electrodes **231** of the pin plate **230**. The power may be current and voltage supplied for cell defect inspection and pre-charging.

[0063] The pin plate **230** may contact the lower surface of the power unit **202**. The pin plate **230** may extend downwards from the lower surface of the power unit **202**. The plurality of electrodes **231** may be exposed and arranged on bottom of the pin plate **230**.

[0064] The bottom plate **240** may be spaced a predetermined distance apart from the pin plate **230**. The jig unit **250** may be disposed on the upper surface of the bottom plate **240**. The jig unit **250** may place, support and align the tray **30**.

[0065] The up/down movement unit **260** is a device that connects the bottom plate **240** to the pin plate **230**. The up/down movement unit **260** may be configured to decrease or increase the distance between the bottom plate **240** and the pin plate **230**. For example, with the bottom plate **240** placed in a fixed state, as the pin plate **230** is moved down by the up/down movement unit **260**, the distance between the bottom plate **240** and the pin plate **230** may decrease. On the contrary, with the pin plate **230** placed in a fixed state, as the bottom plate **240** is moved up by the up/down movement unit **260**, the distance between the bottom plate **240** and the pin plate **230** may decrease.

[0066] Accordingly, it may be possible to form an improved structure of placing the assembled battery cells **20** transported in the two trays **30** and simultaneously carrying the plurality of trays **30** to and from the cell inspection unit **200**, thereby improving the spatial density and the production efficiency of the entire battery cell inspection system.

[0067] Additionally, the cell inspection unit **200** that performs cell defect inspection and pre-charging may simultaneously inspect the battery cells **20** loaded on the two trays **30**, thereby reducing the space occupied by the inspection machine and improving the space efficiency of the battery inspection system. That is, since the first inspection device **210** and the second inspection device **220** are vertically stacked in two levels on top of each other, it requires a space as large as the first inspection device **210** in the horizontal direction, so it may be possible to reduce the equipment space of the cell inspection unit **200** and the entire battery inspection system.

[0068] The shuttle **300** may be a device that moves between the cell inspection unit **200** and the transport unit **100**. The shuttle **300** may carry the tray **30** to the cell inspection unit **200** or take the tray **30** from the cell inspection unit **200**. The shuttle **300** may convey the plurality of trays **30** when performing an operation of carrying to the cell inspection unit **200** once or an operation of taking from the cell inspection unit **200** once.

[0069] Additionally, the shuttle **300** may be multiple levels corresponding to the transport unit **100** and the cell inspection unit **200**, and may convey at least one tray **30** for each level (storey) of the shuttle **300**. As shown in FIG. 2, the shuttle **300** may be two levels corresponding to the two levels of the transport unit **100**, and may convey at least one tray **30** for each of the two levels.

[0070] By the above-described exemplary configuration, it may be possible to form an improved structure of receiving two assembled battery cells **20** transported and simultaneously carrying two trays **30** to and from the cell inspection unit **200**, thereby improving the spatial density and the production efficiency of the entire battery inspection system.

[0071] FIG. 5 is a front view of the cell inspection unit of the battery inspection system according

to an embodiment of the present disclosure, and FIGS. 6 and 7 are top views of the bottom plate and the jig unit of the cell inspection unit of the battery inspection system according to an embodiment of the present disclosure.

[0072] The cell inspection unit **200** of the battery cell inspection system according to this embodiment will be described below in detail with reference to FIGS. 5 to 7 and FIG. 4.

[0073] Referring to FIGS. 5 and 6, the cell inspection unit **200** may include the first inspection device **210** disposed at the height corresponding to the first transport conveyor **110**, and the second inspection device **220** placed on the first inspection device **210** and disposed at the height corresponding to the second transport conveyor **120**.

[0074] Each of the first inspection device **210** and the second inspection device **220** may include the pin plate **230** disposed in the upper part of the body and having the plurality of exposed electrodes **231**; the bottom plate **240** spaced the predetermined distance apart from the pin plate **230**; and the up/down movement unit **260** to move the bottom plate **240** up/down to the pin plate **230**.

[0075] Since the first inspection device **210** and the second inspection device **220** have the same structure, only the first inspection device **210** will be described in detail below.

[0076] The pin plate **230** may be disposed in the lower area of the frame **201**. The pin plate **230** may be connected to the power unit **202** located thereon, and may extend downwards from the power unit **202**. The plurality of electrodes **231** may be exposed and arranged on the bottom of the pin plate **230**. Each of the plurality of electrodes **231** may correspond to each of the battery cells **20**. Additionally, the electrodes **231**, each corresponding to each battery cell **20**, may be an assembly of one or more electrodes **231**. For example, the assembly of electrodes **231** may include positive and negative electrodes **231** and a temperature sensor, and for convenience, only one electrode **231** is shown in the drawing.

[0077] The bottom plate **240** may be spaced the predetermined distance apart from the pin plate **230**. The bottom plate **240** may be configured to move up to the pin plate **230** or down from the pin plate **230**. The jig unit **250** may be disposed on the bottom plate **240**. Additionally, a fixing block **263** may be placed on two sides of the bottom plate **240**.

[0078] The up/down movement unit **260** may include a cylinder **261**, a rod **262** having one end connected to the cylinder **261** and the other end coupled to the bottom plate **240** and the fixing block **263** coupled to the bottom plate **240** and configured to fix the other end of the rod **262**.

[0079] The rod **262** may have one end connected to the cylinder **261** and the other end coupled to the bottom plate **240**. The fixing block **263** may be disposed in the bottom plate **240** and configured to fix the other end of the rod **262**. Through the up/down movement unit **260**, the distance between the bottom plate **240** and the pin plate **230** may decrease or increase. In this embodiment, by the operation of the cylinder **261**, the rod **262** may be extended out and the fixing block **263** fixed to the other end may be moved up/down. Accordingly, with respect to the pin plate **230**, the bottom plate **240** moves up/down.

[0080] Additionally, the first inspection device **210** may further include the jig unit **250** disposed in the bottom plate **240**. The jig unit **250** may support and align the tray **30**. Referring to FIGS. 5 and 6, the jig unit **250** may include a jig plate **251** on which the tray **30** is seated, a ratchet fastener **252** configured to fix two sides of the tray **30** seated on the jig plate **251** when rotated, and a stopper **253** configured to limit movements in a direction of carrying the tray **30** into the jig plate **251**.

[0081] The jig plate **251** may be disposed on the bottom plate **240** and spaced a predetermined distance apart from the bottom plate **240**. The tray **30** may be seated on the jig plate **251**.

[0082] The ratchet fastener **252** may be disposed on two sides of the jig plate **251**. The ratchet fastener **252** may align the tray **30** in position when the tray **30** enters the first inspection device **210**. When the tray **30** is seated on the jig plate **251**, the ratchet fastener **252** may rotate around a rotation axis on one side and an alignment roller **252a** may press the two sides of the tray **30** as shown in FIG. 7. Accordingly, alignment of the tray **30** in the left-right direction may be achieved.

[0083] The stopper **253** may be disposed in the upper area of the jig plate **251** as shown in FIGS. **6** and **7**. The stopper **253** may limit movements in the direction of carrying the tray **30** into the first inspection device **210**. In the drawing, the direction of carrying the tray **30** faces upward, and the stopper **253** is disposed in the upper area of the jig plate **251**, so it may be possible to hold the top of the tray **30** being carried, thereby achieving alignment of the tray **30** in the up-down direction. [0084] By the ratchet fastener **252** and the stopper **253** configured as described above, it may be possible to make it easy to align the tray **30** in position on the jig plate **251**, thereby enabling the first inspection device **210** to easily perform pre-charging and cell defect inspection of the battery cells **20**.

[0085] FIG. **8** is a perspective view of the shuttle of the battery inspection system according to an embodiment of the present disclosure, and FIGS. **9** and **10** are diagrams showing the tray lifted by a fork for gripping **360** and a protruding portion **361** of the shuttle of the battery inspection system according to an embodiment of the present disclosure.

[0086] Referring primarily to FIG. **8**, the shuttle **300** may be two levels corresponding to the two levels of the transport unit **100** and the cell inspection unit **200**. To this end, the shuttle **300** may include a body **310** having a moving wheel **320**; a first carrying unit **330** disposed at a height corresponding to the first transport conveyor **110** and the first inspection device **210**; and a second carrying unit **340** disposed at a height corresponding to the second transport conveyor **120** and the second inspection device **220**.

[0087] The body **310** may have the moving wheel **320** to help the shuttle **300** to move in at least one direction.

[0088] The first carrying unit **330** and the second carrying unit **340** have the same structure, and the structure of the second carrying unit **340** will be described below. Referring primarily to FIG. **8**, the second carrying unit **340** may include a pair of guide rails **350** spaced apart from each other by the width of the tray **30**; the fork for gripping **360** disposed slidably movably along the guide rails **350** and configured to grip two sides of the tray **30**; and a fork driving unit **362** to drive the fork for gripping **360**.

[0089] The second carrying unit **340** may include a support plate, and the guide rails **350** may be coupled to the support plate movably upward/downward. The support plate may include a rail lift **351**.

[0090] The guide rails **350** may be spaced apart from each other by the width of the tray **30**. One end of the guide rails **350** may extend from one side beyond the width (the width in the Y axis direction) of the shuttle **300**. The guide rails **350** may be configured to guide the slidable movement of the fork for gripping **360** during the movement of the fork for gripping **360**, and support an external force (the weight of the tray **30**) applied to the fork for gripping **360**.

[0091] The fork for gripping **360** is the component that grips and conveys the tray **30**. The fork for gripping **360** may be coupled to the guide rails **350** slidably movably along the guide rails **350**. The fork for gripping **360** may move from the body **310** in a direction perpendicular to the movement direction of the shuttle **300**.

[0092] The fork for gripping **360** may have the protruding portion **361** on one surface thereof, and the protruding portion **361** may be fit-coupled to a stopper step **31** of the tray **30**. Referring to FIGS. **9** and **10**, the protruding portion **361** may slightly lift the tray **30** by upward movement from the stopper step **31** of the tray **30**.

[0093] The second carrying unit **340** may convey one tray **30** when performing an operation of carrying to or from the cell inspection unit **200** once. Additionally, as shown in FIG. **8**, likewise, the first carrying unit **330** disposed vertically below the second carrying unit **340** may convey one tray **30**. That is, as shown in FIG. **1**, the shuttle **300** may simultaneously carry two trays **30** into the cell inspection unit **200**, each one tray **30** for each of the first carrying unit **330** and the second carrying unit **340** in the operation of carrying once, and may take the two trays **30** from the cell inspection unit **200** by the reverse operation. The shuttle **300** may move in parallel to the lengthwise

arrangement of the two cell inspection units **200**.

[0094] Through the shuttle **300** configured as described above, it may be possible to form a structure of placing the assembled battery cells **20** transported in the two trays **30** and simultaneously carrying the plurality of trays **30** to and from the cell inspection unit **200**.

Accordingly, it may be possible to improve the spatial density and the production efficiency of the entire battery inspection system.

[0095] Additionally, since each of the first transport conveyor **110** and the second transport conveyor **120** transports each tray **30** and the total of two trays **30** are simultaneously carried to and from the first inspection device **210** and the second inspection device **220**, it may be possible to reduce the time required to carry the battery cells **20** and perform the pre-charging and operation inspection process of the battery cells **20** quickly and efficiently.

[0096] FIGS. **11** to **13** are diagrams showing an example in which the cell inspection unit of the battery inspection system according to an embodiment of the present disclosure accommodates the trays from the transport unit and performs cell defect inspection and pre-charging.

[0097] Hereinafter, the inspection process of the battery cells **20** through the battery cell inspection system according to this embodiment will be described with reference to FIGS. **1** to **13**.

[0098] First, as shown in FIG. **2**, each one tray **30** is transported from the assembly unit **5** to the first transport conveyor **110** and the second transport conveyor **120** of the transport unit **100**. The tray from the assembly unit **5** may be temporarily loaded on the station **11** (see FIG. **1**).

[0099] Subsequently, the shuttle **300** approaches the transport unit **100** and the fork for gripping **360** of each of the first carrying unit **330** and the second carrying unit **340** grips the tray **30** on each of the first transport conveyor **110** and the second transport conveyor **120**. Specifically, the fork for gripping **360** in FIG. **8** moves in a direction perpendicular to the movement direction of the shuttle **300** from the body **310** by the operation of the fork driving unit **362**. The fork for gripping **360** slidably moves in the Y axis direction, and although not shown, approaches the tray **30** on the two levels of the transport unit **100**.

[0100] Additionally, as shown in FIGS. **9** and **10**, the rail lift **351** pushes up the guide rails **350** and accordingly the protruding portion **361** of the fork for gripping **360** is fitted into the stopper step **31** of the tray **30** and lifts up the tray **30**. Additionally, as shown in FIG. **11**, the fork for gripping **360** moves to the body **310** again and is seated in the shuttle **300**. The first carrying unit **330** in the seated state is shown. This operation may be performed by the first carrying unit **330** and the second carrying unit **340** at the same time, and for convenience of description, only the operation of the second carrying unit **340** is shown.

[0101] Subsequently, the shuttle **300** moves to the empty cell inspection unit **200** as shown in FIG. **2**.

[0102] Subsequently, the shuttle **300** carries the tray **30** into the cell inspection unit **200**, and specifically, as shown in FIG. **12**, the fork for gripping **360** of the second carrying unit **340** slidably moves to the second inspection device **220** and moves the tray **30** being gripped to the upper surface of the jig unit **250** of the second inspection device **220**. During the movement of the tray **30**, the stopper **253** in FIG. **7** is fitted into an upper region of the tray **30** to limit the upward movement of the tray **30**. Additionally, as shown in FIG. **8**, the rail lift **351** moves down to seat the tray **30** on the upper surface of the jig unit **250**.

[0103] In this instance, as shown in FIGS. **6** and **7**, the ratchet fastener **252** slightly rotates and the alignment roller **252a** presses the two sides of the tray **30**, completing the alignment of the tray **30** in the left-right direction. The process of carrying the tray **30** into the cell inspection unit **200** may be performed by not only the second inspection device **220** but also the first inspection device **210**.

[0104] Subsequently, the cell inspection unit **200** performs pre-charging and cell defect inspection of the battery cells **20**. Referring to FIG. **13**, the rod **262** is reduced by the operation of the cylinder **261** and the fixing block **263** fixed to the other end moves up, and accordingly, with respect to the pin plate **230**, the bottom plate **240** moves up. The battery cells **20** electrically contact the

electrodes **231** on the pin plate **230**. Subsequently, the pre-charging and cell defect inspection of the battery cells **20** is performed.

[0105] Meanwhile, when the cell inspection unit **200** completes the inspection, the process of taking the tray **30** from the cell inspection unit **200** is performed. The process of taking the tray **30** from the cell inspection unit **200** is performed in the opposite order to the process of carrying the tray **30** into the cell inspection unit **200**, and finally, the tray **30** is taken to the first transport conveyor **110** and the second transport conveyor **120** of the transport unit **100** on the carry-out path, and the trays **30** may be loaded in multilayer to form the loaded item **7**, followed by aging through the aging machine **9** and the post-process such as the charging/discharging process in a sequential order.

[0106] By the above-described exemplary configuration, it may be possible to form a structure of placing the assembled battery cells **20** transported in the two trays **30** and simultaneously carrying the plurality of trays **30** to and from the cell inspection unit **200**. Accordingly, it may be possible to improve the spatial density and the production efficiency of the entire battery inspection system.

[0107] Additionally, the cell inspection unit **200** that performs cell defect inspection and pre-charging may simultaneously inspect the battery cells **20** loaded on the two trays **30**, thereby reducing the space occupied by the inspection machine and improving the space efficiency of the battery inspection system.

[0108] By the above-described exemplary configuration, it may be possible to simultaneously carry two trays **30** on which the battery cells **20** are mounted to and from the cell inspection unit **200**, thereby reducing the inspection time and improving the inspection speed, resulting in reduced takt time of the formation process of the battery cells **20** and increased production efficiency of the battery cells **20**.

[0109] While the present disclosure has been hereinabove described with regard to a limited number of embodiments and drawings, the present disclosure is not limited thereto and it is apparent that a variety of changes and modifications may be made by those skilled in the art within the technical aspect of the present disclosure and the scope of the appended claims and their equivalents.

[0110] The terms indicating directions such as upper, lower, left, right, front and rear are used for convenience of description, but it is obvious to those skilled in the art that the terms may change depending on the position of the stated element or an observer.

Claims

1. A battery cell inspection system, comprising: a transport unit including a plurality of conveyors arranged in multiple levels in a vertical direction, and configured to transport a tray on which an assembled battery cell is mounted; a cell inspection unit corresponding to the multiple levels of the transport unit, and configured to perform pre-charging and cell defect inspection of the assembled battery cell; and a shuttle configured to move between the cell inspection unit and the transport unit to carry the tray into the cell inspection unit or take the tray from the cell inspection unit.
2. The battery cell inspection system according to claim 1, wherein the shuttle is configured to convey a plurality of trays when carrying the plurality of trays to the cell inspection unit or when taking the plurality of trays from the cell inspection unit.
3. The battery cell inspection system according to claim 2, wherein the shuttle comprises multiple levels corresponding to the transport unit and the cell inspection unit, and wherein each level of the multiple levels of the shuttle is configured to convey at least one tray.
4. The battery cell inspection system according to claim 1, wherein the transport unit is disposed on a carry-in path and a carry-out path of the cell inspection unit.
5. The battery cell inspection system according to claim 3, wherein the transport unit includes: a first transport conveyor; and a second transport conveyor disposed a vertical distance from the first

transport conveyor.

- 6.** The battery cell inspection system according to claim 5, wherein the cell inspection unit further includes: a first inspection device disposed at a height corresponding to the first transport conveyor; and a second inspection device placed on the first inspection device and disposed at a height corresponding to the second transport conveyor.
 - 7.** The battery cell inspection system according to claim 6, wherein each of the first inspection device and the second inspection device includes: a pin plate disposed on the tray and having a plurality of exposed electrodes; a bottom plate configured to support the tray and spaced a predetermined distance apart from the pin plate; and an up/down movement unit connecting the bottom plate to the pin plate, and configured to decrease or increase a distance between the bottom plate and the pin plate.
 - 8.** The battery cell inspection system according to claim 7, wherein each of the first inspection device and the second inspection device further includes a jig unit disposed on the bottom plate and configured to support and align the tray.
 - 9.** The battery cell inspection system according to claim 7, wherein the up/down movement unit includes: a cylinder; a rod having an end connected to the cylinder and an opposite end coupled to the bottom plate; and a fixing block coupled to the bottom plate and configured to couple the opposite end of the rod to the bottom plate.
 - 10.** The battery cell inspection system according to claim 8, wherein the jig unit includes: a jig plate configured to receive the tray; and a ratchet fastener configured to fix two sides of the tray received on the jig plate when the ratchet fastener is rotated.
 - 11.** The battery cell inspection system according to claim 10, wherein the jig unit further includes a stopper configured to limit movement in a direction of carrying the tray into the jig plate.
 - 12.** The battery cell inspection system according to claim 6, wherein a plurality of cell inspection units are arranged in a line.
 - 13.** The battery cell inspection system according to claim 12, wherein the shuttle moves in parallel to a lengthwise arrangement of the plurality of cell inspection units.
 - 14.** The battery cell inspection system according to claim 13, wherein the shuttle includes: a body having a moving wheel; a first carrying unit disposed at the height corresponding to the first transport conveyor and the first inspection device; and a second carrying unit disposed at the height corresponding to the second transport conveyor and the second inspection device.
 - 15.** The battery cell inspection system according to claim 14, wherein each of the first carrying unit and the second carrying unit includes: a pair of guide rails spaced apart from each other by a width corresponding to a width of the tray; a fork for gripping that is configured to slide along the pair of guide rails and configured to grip two sides of the tray; and a fork driving unit configured to drive the fork for gripping.
 - 16.** The battery cell inspection system according to claim 15, wherein the fork for gripping moves from the body in a direction perpendicular to a movement direction of the shuttle.
 - 17.** The battery cell inspection system according to claim 1, wherein the cell defect inspection comprises causing a predetermined amount of current to flow in the assembled battery cell, detecting a voltage change of the assembled battery cell, and determining a presence or absence of a defect.
 - 18.** The battery cell inspection system according to claim 1, wherein the assembled battery cell includes an electrode assembly including a negative electrode plate, and wherein the pre-charging of the assembled battery cell comprises charging the assembled battery cell in a predetermined amount to form an electric charge layer on the negative electrode plate to prevent formation of an electron movement channel by oxidation of metal impurities.
 - 19.** The battery cell inspection system according to claim 1, wherein the assembled battery cell is a cylindrical battery.
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